



Weighted Heisenberg–Pauli–Weyl uncertainty principles for the linear canonical transform

Qiang Feng^a, Bing-Zhao Li^{b,*}, John-Michael Rassias^c

^aSchool of Mathematics and Computer Science, Yanan University, Yanan, Shaanxi 716000, China

^bSchool of Mathematics and Statistics, Beijing Institute of Technology, Beijing 102488, China

^cPedagogical Department, National and Capodistrian University, Athens, Greece

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ABSTRACT

The classical uncertainty principle plays an important role in quantum mechanics, signal processing and applied mathematics. With the development of novel signal processing methods, the research of the related uncertainty principles has gradually been one of the most hottest research topics in modern signal processing community. In this paper, the weighted Heisenberg–Pauli–Weyl uncertainty principles for the linear canonical transform (LCT) have been investigated in detail. Firstly, the Plancherel–Parseval–Rayleigh identities associated with the LCT are derived. Secondly, the weighted Heisenberg–Pauli–Weyl uncertainty principles in the LCT domain are investigated based on the derived identities. The signals that can achieve the lower bound of the uncertainty principle are also obtained. The classical Heisenberg uncertainty principles in the Fourier transform (FT) domain are shown to be special cases of our achieved results. Thirdly, examples are provided to show that our weighted Heisenberg–Pauli–Weyl uncertainty principles are sharper than those in the existing literature. Finally, applications of the derived results in time frequency resolution analysis and signal energy concentrations are also analyzed and discussed in detail.

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1. Introduction

The classical Heisenberg uncertainty principle was established by Heisenberg [1], who introduced the fundamental problem in quantum mechanics: *The position and the momentum of an electron within an atom cannot be both determined explicitly but only in a probabilistic sense under a certain “uncertainty”*. In signal processing community, the classical Heisenberg uncertainty principle for the Fourier transform (FT) shows that the product of the duration and bandwidth of a signal $f(x)$ possesses a lower bound. The mathematical representation of this uncertainty principle was provided by Pauli and Weyl [2] as follows

$$\sigma_x^2 \cdot \sigma_\xi^2 \geq \frac{E_{|f|^2}^2}{16\pi^2}, \quad (1)$$

where $E_{|f|^2}$ and $F_f(\xi)$ of a signal $f(x)$ is given by

$$E_{|f|^2} = \|f\|_2^2 = \int_{\mathbb{R}} |f(x)|^2 dx, \quad (2)$$

and

$$F_f(\xi) = \int_{\mathbb{R}} f(x) e^{-i2\pi\xi x} dx. \quad (3)$$

respectively. If $F_f(\xi)$ is also in $L^1(\mathbb{R})$, then the inversion formula holds [3], that is

$$f(x) = \int_{\mathbb{R}} F_f(\xi) e^{i2\pi\xi x} d\xi. \quad (4)$$

Through a density argument, both the FT and its inverse can be extended to $L^2(\mathbb{R})$. When using the formulas (3) and (4) for $L^2(\mathbb{R})$ functions, we keep in mind that the convergence of the integrals is in the $L^2(\mathbb{R})$ sense. Eq. (1) is called the Heisenberg–Pauli–Weyl uncertainty principle [1–5] associated with the classical FT. Many works have been dedicated to establishing this kind of the uncertainty principle [6–11] and other generalized versions [12–15].

The classical Heisenberg–Pauli–Weyl uncertainty principle is a fundamental result in signal analysis, since the product of time spread and spectrum bandwidth of FT cannot both be made arbitrarily small. In recent years, many kinds of these results about the uncertainty principle [16–25] has been extended to the LCT.

The LCT [26–30] is the linear integral transform with three free parameters, which has been proved to be one of the most powerful tools in the signal processing community and has more flexibility

* Corresponding author.

E-mail address: li_bingzhao@bit.edu.cn (B.-Z. Li).

Notations and nomenclature

FT	Fourier transform
FRFT	Fractional Fourier transform
LCT	Linear canonical transform
2D FRT	Two dimensional Fresnel Transform
2D FT	Two dimensional Fourier transform
2D FRFT	Two dimensional fractional Fourier transform
2D LCT	Two dimensional linear canonical transform
2D NSLCT	Two dimensional nonseparable linear canonical transform

than the one free parameter of the FRFT and the zero free parameter of the FT. It is more suitable for non-stationary signal processing because it can realize the extension and rotation of a signal in time frequency plane. Many transforms such as the FT [3], the FRFT [26] and the FRT [4] are all its special cases.

The classical concepts and methods related to the FT and the FRFT have been extended to LCT domain, for example, the filtering [31–33], sampling [34–39], and spectral analysis [40–44], and so on. Among them, the research of the uncertainty principles associated with the LCT has become one of the most interesting research topics during the last two decades. The novel uncertainty principles for the LCT have been derived (summarized in Tables 1 and 2). It can be seen from these results that we can obtain the different lower uncertainty bounds for different kind of transforms, and the research results can further guide theoretical research and engineering applications in related fields. Therefore, it is of great significance to investigate the uncertainty principles in different transform domains.

From Tables 1 and 2, we can conclude that all of the existing results about the uncertainty principle related to LCT are based on the second order statistic moment of signal. However, we will obtain suboptimal results for non-stationary signal analysis if we still use second order statistic properties because higher-order moments contain more useful information than that of the variances. It is therefore interesting and worthwhile to investigate the relationship among the higher order statistics moments of non-stationary signals in LCT domain.

To the best of our knowledge, the following two aspects of uncertainty principles related to the LCT have not reported: the first one relates to the higher-order moments of signal, and the other one relates to the weight of the uncertainty principle.

Table 1
Heisenberg–Pauli–Weyl uncertainty principles for LCT.

Lower bound of uncertainty product	The function that can minimize the product	Type	Reference
$\sigma_x^2 \sigma_{\xi,A}^2 \geq \frac{b^2}{4}$	$re^{i\omega_0 x} e^{-i\frac{1}{2}abx^2} e^{-\frac{(x-x_0)^2}{4\sigma^2}}$	C	[16]
$\sigma_x^2 \sigma_{\xi,A}^2 \geq \frac{b^2}{4} + (a\Delta x^2)^2$	$re^{-\frac{(x-x_0)^2}{4\sigma^2}}$	R	[16]
$\sigma_x^2 \sigma_{\xi,A}^2 \geq \frac{b^2}{4} + (a\Delta x^2 + b\text{Cov}_{x,w})^2$	$(\frac{1}{2\pi\Delta x^2})^{\frac{1}{2}} e^{-\frac{(x-x_0)^2}{4\Delta x^2}} e^{i(-\frac{1}{2}x^2 + \Omega x + \phi)}$	A	[19]
$\sigma_{\xi,A_1}^2 \sigma_{\xi,A_2}^2 \geq \frac{(a_1 b_2 - a_2 b_1)^2}{4}$	$(\frac{1}{\pi\Delta x^2})^{\frac{1}{2}} e^{-\frac{(x-x_0)^2}{2\Delta x^2}} e^{i(-\frac{1}{2}x^2 + \Omega x + \phi)}$	C	[18]
$\sigma_{\xi,A_1}^2 \sigma_{\xi,A_2}^2 \geq \frac{(a_1 b_2 - a_2 b_1)^2}{4} + (a_1 a_2 \Delta x^2 + \frac{b_1 b_2}{4\Delta x^2})^2$	$(\frac{1}{\sigma^2})^{\frac{1}{2}} e^{-\frac{x^2}{2\sigma^2}} e^{i(-\frac{1}{2}x^2 + \Omega x + \phi)}$	R	[17,18,20]
$\sigma_{\xi,A_1}^2 \sigma_{\xi,A_2}^2 \geq \frac{(a_1 b_2 - a_2 b_1)^2}{4} + (a_1 a_2 \Delta x^2 + b_1 b_2 \Delta w^2 + (a_1 b_2 + a_2 b_1)\text{Cov}_{x,w})^2$	$(\frac{1}{2\pi\Delta x^2})^{\frac{1}{2}} e^{-\frac{(x-x_0)^2}{4\Delta x^2}} e^{i(-\frac{1}{2}x^2 + \Omega x + \phi)}$	A	[19]
$\sigma_{\xi,A_1}^2 \sigma_{\xi,A_2}^2 \geq \frac{(a_1 b_2 - a_2 b_1)^2}{4} + (a_1 a_2 \Delta x^2 + b_1 b_2 \Delta w^2 + b_1 b_2 K_1 + (a_1 b_2 + a_2 b_1)K_2)^2$	$K_2 e^{-i\frac{(K_1 \phi_2 - a_1) x^2}{2(K_1 b_2 - b_1)}}$	C	[21]
$\sigma_{\xi,A_1}^2 \sigma_{\xi,A_2}^2 \geq (a_1 b_2 - a_2 b_1)^2 \cdot (\frac{1}{4} + \text{Cov}_{x,w}^2 - \text{Cov}_{x,w}^2) + (a_1 a_2 \Delta x^2 + b_1 b_2 \Delta w^2 + (a_1 b_2 + a_2 b_1)\text{Cov}_{x,w})^2$	$e^{-\frac{1}{2}(x - \langle x \rangle)^2 + F_1} e^{i(\frac{1}{2}(x - \langle x \rangle)^2 + \langle w \rangle x + F_2)}$	C	[22]
$\sigma_{\xi,A_1}^2 \sigma_{\xi,A_2}^2 \geq \{\text{Max}\{U_{\text{Cov}}, V_{\text{Cov}}\}, \text{ or } \text{Max}\{U_{\text{Cov}}, V_{\text{Cov}}\}\}$	$(\frac{1}{\pi})^{\frac{1}{2}} e^{-\frac{1}{2}(x-x_0)^2} e^{i(\frac{1}{2}(x-x_0)^2 + \omega_0 x + F_1)}$	C	[23]
$ \Omega_1 \Omega_2 \geq (1 - \epsilon_1 - \epsilon_2)^2 2\pi T$	Not given	C	[24]
$\sigma_{\omega,x}^2 \sigma_{\xi,A}^2 \geq \frac{b^2}{4} E_{p,f}^2$	$e^{i\frac{b}{2}x} e^{-i\frac{b}{2}x^2} H(x)$	C	Present Study

Note 1. $p = 1, 2, \dots$, $A = (a, b, c, d)$, $A_1 = (a_1, b_1, c_1, d_1)$, $A_2 = (a_2, b_2, c_2, d_2)$. $r, r_1, r_2, k, l, K_1, K_2, \Omega, \phi, \epsilon$ are all arbitrary real constants.
2. “A” denotes an arbitrary signal, “R” denotes a real signal, and “C” denotes a complex signal.

The purpose of this paper is to investigate the weighted version uncertainty principles, and obtain low bounds of weighted Heisenberg–Pauli–Weyl uncertainty principles based on the higher-order statistic moment of non-stationary signal, which contain more parameters, so we can control or adjust parameters to make the uncertainty principles more sharper and thus much stronger than the existing results. We first establish the Plancherel–Parseval–Rayleigh identities related to LCT and 2D LCT, then, we investigate the $2p$ th weighted moment Heisenberg–Pauli–Weyl uncertainty principles for the LCT and 2D LCT based on the derived identities. Further, we derive lower bounds of weighted version Heisenberg–Pauli–Weyl uncertainty principles. The simulations and examples show that the lower bounds of derived weighted uncertainty principles are tighter than those of the latest results [18,21,22]. Finally, time frequency resolution analysis and signal energy concentrations in signal processing are also discussed based on our achieved results. Table 3 shows the research status between the existing results and the obtained results in this paper.

The rest of this paper is organized as follows. Section 2 is preliminaries. In Section 3, we obtain the Plancherel–Parseval–Rayleigh identities for LCT and 2D LCT. The weighted Heisenberg–Pauli–Weyl uncertainty principles for the LCTs are investigated and derived in Section 4. The potential applications are discussed in Section 5. Finally, conclusions are summarized in Section 6.

The following are the transform results used in this paper.

- Transform results of $f(x)$ for LCT, FRFT, FT and FRT: $L_f^A(\xi), F_f^\theta(\xi), F_f(\xi), FR_f(\xi)$.

- Transform results of $f(x, y)$ for 2DNSLCT, 2D LCT, 2D FRFT and 2D FT: $F_f^M(\xi, \eta), F_f^{\tilde{A}, \tilde{B}}(\xi, \eta), F_f^{\alpha, \beta}(\xi, \eta)$ and $F_f(\xi, \eta)$.

The following are the notations of moment used in this paper.

- $\sigma_x^{2p} = \int_{\mathbb{R}} \omega^2(x) (x - x_m)^{2p} |f(x)|^2 dx$, $\sigma_x^{2p} = \int_{\mathbb{R}} (x - x_m)^{2p} |f(x)|^2 dx$.
- $\sigma_{\xi,A}^{2p} = \int_{\mathbb{R}} (\xi - \xi_m)^{2p} |L_f^A(\xi)|^2 d\xi$.
- $\sigma_{\xi,\theta}^{2p} = \int_{\mathbb{R}} (\xi - \xi_m)^{2p} |F_f^\theta(\xi)|^2 d\xi$.
- $\sigma_{\xi,r}^{2p} = \int_{\mathbb{R}} (\xi - \xi_m)^{2p} |FR_f(\xi)|^2 d\xi$.
- $\sigma_{\xi}^{2p} = \int_{\mathbb{R}} (\xi - \xi_m)^{2p} |F_f(\xi)|^2 d\xi$.
- $\sigma_{\omega,x,y}^{2p} = \int_{\mathbb{R}^2} (\omega_{1,p}^2(x, y) + \omega_{2,p}^2(x, y)) |f(x, y)|^2 dx dy$, $\sigma_{x,y}^{2p} = \int_{\mathbb{R}^2} ((x - x_m)^{2p} + (y - y_m)^{2p}) |f(x, y)|^2 dx dy$, where $\omega_{1,p}(x, y) = (x - x_m)^p \omega(x, y)$ and $\omega_{2,p}(x, y) = (y - y_m)^p \omega(x, y)$.
- $\sigma_{\xi,\eta,M}^{2p} = \int_{\mathbb{R}^2} ((\xi - \xi_m)^{2p} + (\eta - \eta_m)^{2p}) |F_f^M(\xi, \eta)|^2 d\xi d\eta$.
- $\sigma_{\xi,\eta,\tilde{A},\tilde{B}}^{2p} = \int_{\mathbb{R}^2} ((\xi - \xi_m)^{2p} + (\eta - \eta_m)^{2p}) |F_f^{\tilde{A}, \tilde{B}}(\xi, \eta)|^2 d\xi d\eta$.

Table 2

Heisenberg–Pauli–Weyl uncertainty principles for 2D operators.

Transform domain	Lower bound of uncertainty product	The function that can minimize the product	Type	Reference
2D FRT	$\sigma_{x,y}^2, \sigma_{\xi,\eta}^2 \geq \frac{\lambda^2 z^2}{4\pi^2}$	$Ke^{-i\pi \frac{x^2+y^2}{\lambda^2}} e^{-\pi \frac{x^2+y^2}{\lambda^2}}$	A	[25]
2D FT	$\sigma_{x,y}^2 \sigma_{\xi,\eta}^2 \geq 1$	$Ke^{-\frac{x^2+y^2}{2}}$	A	[25]
2D FRFT	$\sigma_{x,y}^2 \sigma_{\xi,\eta,\alpha,\beta}^2 \geq \frac{(\sin \alpha + \sin \beta)^2}{4}$	$Ke^{-\frac{x^2}{2} \frac{ \cos \alpha + \cos \beta }{2}} \cdot e^{-i \frac{x^2 \cos \alpha + y^2 \cos \beta}{2}}$	A	[25]
2D LCT	$\sigma_{x,y}^2 \sigma_{\xi,\eta,\tilde{A},\tilde{B}}^2 \geq \frac{(b_1 + b_2)^2}{4}$	$Ke^{-(\frac{x^2}{2b_1} + \frac{y^2}{2b_2})} \cdot e^{-i(\frac{a_1^2}{2b_1} + \frac{a_2^2}{2b_2})}$	A	[25]
2D NSLCT	$\sigma_{x,y}^2 \sigma_{\xi,\eta,M}^2 \geq \max((b_{11} + b_{22})^2 + (b_{12} - b_{21})^2, (b_{11} - b_{22})^2 + (b_{12} + b_{21})^2)/4$	$Ke^{-\frac{(x_1 \cos \theta - y_1 \sin \theta)^2 / 2}{2}} \cdot e^{-\frac{(x_1 \sin \theta + y_1 \cos \theta)^2 / 2}{2}} \cdot e^{-i \frac{(p_1 x_1^2 + p_2 x_1 y_1 + p_3 y_1^2)}{2 b }}$	A	[25]
weighted 2D LCT	$(\sigma_{\omega,x,y}^2 \sigma_{\xi,\eta,\tilde{A},\tilde{B}}^2 \geq \frac{b_1^{2p}}{4} (E_{1,x}^2 + E_{2,x}^2) + \frac{b_2^{2p}}{4} (E_{1,y}^2 + E_{2,y}^2))$	$ce^{i(\frac{\xi_1 x}{b_1} + \frac{\xi_2 y}{b_2})} \cdot e^{-i(\frac{a_1 x^2}{2b_1} + \frac{a_2 y^2}{2b_2})} H(x, y)$	C	Present Study

Note: “A” denotes an arbitrary signal, “C” denotes a complex signal.

Table 3

Comparison of 2th Heisenberg–Pauli–Weyl uncertainty principles with 2pth weighted cases.

Transform domain	2th moment HPWUP	2pth weighted moment HPWUP
FT	✓	[16]
2D FT	✓	Present Study
FRFT	✓	Present Study
2D FRFT	✓	Present Study
LCT	✓	Present Study
2D LCT	✓	Present Study
2D NSLCT	✓	×

Note: 1. “✓” denotes results that have been studied; “×” denotes results that have not been studied.

2. HPWUP denotes the Heisenberg–Pauli–Weyl uncertainty principles.

- $\sigma_{\xi,\eta,\alpha,\beta}^{2p} = \int_{\mathbb{R}^2} ((\xi - \xi_m)^{2p} + (\eta - \eta_m)^{2p}) |F_f^{\alpha,\beta}(\xi, \eta)|^2 d\xi d\eta$,
- $\sigma_{\xi,\eta}^{2p} = \int_{\mathbb{R}^2} ((\xi - \xi_m)^{2p} + (\eta - \eta_m)^{2p}) |F_f(\xi, \eta)|^2 d\xi d\eta$.

2. Preliminaries

2.1. Linear canonical transform

Definition 1. The LCT of $f(x) \in L^2(\mathbb{R})$ is defined as follows [26–30]

$$L_f^A(\xi) = \begin{cases} \int_{\mathbb{R}} f(x) K_A(x, \xi) dx, & b \neq 0 \\ \sqrt{d} e^{i\frac{cd}{2} \xi^2} f(d\xi), & b = 0 \end{cases} \quad (5)$$

where $K_A(x, \xi) = \frac{1}{\sqrt{i2\pi b}} e^{i\frac{ax^2 - 2\xi x + d\xi^2}{2b}}$ is the transform kernel of LCT.

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ denotes the real matricial parameters of LCT satisfying $ad - bc = 1$. For more detailed definitions and properties of LCT, one can refer to [26–28].

When $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, LCT reduces to FRFT [26], which is defined as follows

$$F_f^\theta(\xi) = \int_{\mathbb{R}} f(x) K^\theta(x, \xi) dx, \quad (6)$$

where $K^\theta(x, \xi) = \frac{1}{\sqrt{i2\pi \sin \theta}} e^{i\frac{1}{2}(\xi^2 + x^2) \cot \theta - i\xi x \csc \theta}$, $\theta \in \mathbb{R}$, $\theta \neq n\pi$, $n \in \mathbb{Z}$.

When $A = \begin{bmatrix} 0 & 1 \\ -2\pi & 0 \end{bmatrix}$, LCT reduces to FT [3].

Definition 2. The 2D LCT of $f(x, y) \in L^2(\mathbb{R}^2)$ is defined as [28,45–47]

$$F_f^{\tilde{A}, \tilde{B}}(\xi, \eta)$$

$$= \begin{cases} \int_{\mathbb{R}^2} f(x, y) K_{\tilde{A}, \tilde{B}}(\xi, \eta, x, y) dx dy, & b_1 b_2 \neq 0, |\tilde{A}| = |\tilde{B}| = 1 \\ \sqrt{d_1 d_2} e^{i\frac{1}{2}(c_1 d_1 \xi^2 + c_2 d_2 \eta^2)} f(d_1 \xi, d_2 \eta), & b_1^2 + b_2^2 = 0 \end{cases} \quad (7)$$

where the transform kernel is given by

$$K_{\tilde{A}, \tilde{B}}(\xi, \eta, x, y) = K_{\tilde{A}}(\xi, x) K_{\tilde{B}}(\eta, y) \quad (8)$$

$$K_{\tilde{A}}(\xi, x) = C_{\tilde{A}} e^{i(\frac{d_1 \xi^2}{2b_1} - \frac{\xi x}{b_1} + \frac{a_1 x^2}{2b_1})}, \quad C_{\tilde{A}} = \sqrt{\frac{1}{i2\pi b_1}}, \quad \tilde{A} = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix},$$

$$K_{\tilde{B}}(\eta, y) = C_{\tilde{B}} e^{i(\frac{d_2 \eta^2}{2b_2} - \frac{\eta y}{b_2} + \frac{a_2 y^2}{2b_2})}, \quad C_{\tilde{B}} = \sqrt{\frac{1}{i2\pi b_2}}, \quad \tilde{B} = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}.$$

$\tilde{A}$ and $\tilde{B}$ are real parameter matrices. When

$$\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} = \begin{bmatrix} \cos \theta_1 & \sin \theta_1 \\ -\sin \theta_1 & \cos \theta_1 \end{bmatrix}, \quad \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} = \begin{bmatrix} \cos \theta_2 & \sin \theta_2 \\ -\sin \theta_2 & \cos \theta_2 \end{bmatrix}, \quad (9)$$

2D LCT reduces to 2D FRFT [26,28,48]. When $a_1 = a_2 = d_1 = d_2 = 0$, $b_1 = b_2 = \frac{1}{2\pi}$, and $c_1 = c_2 = -2\pi$ in (9), 2D LCT reduces to 2D FT [29,48].

2.2. 2pth weighted uncertainty principle for FT

Different from the other kind of classical uncertainty principles, Rassias investigated the 2pth weighted moment Heisenberg–Pauli–Weyl uncertainty principle for FT firstly in order to sharpen and refine the uncertainty principle inequality [49,52]. We review some of his important results that will be used in the following sections of this paper.

Lemma 1. Assume that $f(x) \in L^2(\mathbb{R})$ is a complex signal and $\omega: \mathbb{R} \rightarrow \mathbb{R}$ is a real weight signal. Let $\omega_p(x) = (x - x_m)^p \omega(x)$, where $x_m, \xi_m \in \mathbb{R}$, $p \in \mathbb{N}_0 = \mathbb{N} \setminus \{0\}$. The 2pth weighted moment Heisenberg–Pauli–Weyl uncertainty principle is given by [49,50,51,52]

$$\sigma_{\omega,x}^{2p} \cdot \sigma_{\xi}^{2p} \geq \frac{1}{2^{2(p+1)} \pi^{2p}} E_{p,f}^2, \quad (10)$$

if the following two conditions hold

$$\lim_{|x| \rightarrow \infty} \omega_p^{(r)}(x) (|f^{(l)}(x)|^2)^{p-2q-r-1} = 0, \quad (11)$$

$$\lim_{|x| \rightarrow \infty} \omega_p^{(r)}(x) [\operatorname{Re}(r_{qkj} f^{(k)}(x) \overline{f^{(j)}(x)})]^{p-2q-r-1} = 0, \quad (12)$$

where

$$E_{p,f} = \sum_{q=0}^{\lfloor \frac{p}{2} \rfloor} D_q \bar{D}_q, \quad |E_{p,f}| < \infty, \quad (13)$$

and

$$D_q = (-1)^q C_{p-q}^q \frac{p}{p-q} \quad (14)$$

$$\bar{D}_q = \sum_{l=0}^q G_{ql} I_{ql} + 2 \sum_{0 \leq k < j \leq q} H_{qkj} I_{qkj}, \quad (15)$$

$$I_{ql} = (-1)^{(p-2q)} \int_{\mathbb{R}} \omega_p^{(p-2q)}(x) |f^{(l)}(x)|^2 dx, \quad (16)$$

$$I_{qkj} = (-1)^{(p-2q)} \int_{\mathbb{R}} \omega_p^{(p-2q)}(x) \operatorname{Re}(r_{qkj} f^{(k)}(x) \overline{f^{(j)}(x)}) dx, \quad (17)$$

$$r_{qkj} = (-1)^{q-\frac{k+j}{2}} \in \{\pm 1, \pm i\}, \quad i = \sqrt{-1}, \quad (18)$$

$$G_{ql} = (C_q^l)^2 \mu^{2(q-l)}, \quad H_{qkj} = (-1)^{q-k} C_q^k C_q^j \mu^{2q-j-k}, \quad (19)$$

where $\mu = 2\pi\xi_m$, $0 \leq r \leq p-2q-1$, $0 \leq l \leq q \leq \lfloor \frac{p}{2} \rfloor$, $0 \leq k < j \leq q \leq \lfloor \frac{p}{2} \rfloor$, $|D_q| < \infty$, $C_q^{(\cdot)}$ and $\overline{(\cdot)}$ denotes the combinatorial and conjugate operation, $\lfloor \frac{k}{2} \rfloor$ is the greatest integer that is not greater than $\frac{k}{2}$, $\operatorname{Re}(\cdot)$ is the real part of $(\cdot)$.

2.3. Rassias propositions

This subsection reviews some propositions derived in [49] by Rassias, and they will be used in this paper.

Proposition 1. If $f: \mathbb{R} \rightarrow \mathbb{C}$ is a complex-valued signal, $f^{(k)}(x) = \frac{d^k}{dx^k} f(x)$. Then,

$$\begin{aligned} f(x) \overline{f^{(k)}(x)} + \overline{f(x)} f^{(k)}(x) &= \sum_{j=0}^{\lfloor \frac{k}{2} \rfloor} (-1)^j C_{k-j}^j \frac{k}{k-j} \frac{d^{k-2j}}{dx^{k-2j}} |f^{(j)}(x)|^2 \\ &= \sum_{j=0}^{\lfloor \frac{k}{2} \rfloor} D_j \frac{d^{k-2j}}{dx^{k-2j}} |f^{(j)}(x)|^2 \end{aligned} \quad (20)$$

holds for $k \in \mathbb{N}_0$, $0 \leq j \leq \lfloor \frac{k}{2} \rfloor$, $j \in \mathbb{N}$.

Proposition 2. If $f: \mathbb{R} \rightarrow \mathbb{C}$ is a complex-valued signal, $f_{\tilde{\alpha}}(x) = e^{\tilde{\alpha}x} f(x)$, where $\tilde{\alpha} = -i2\pi\xi_m$. Then,

$$|f_{\tilde{\alpha}}^{(p)}(x)|^2 = \sum_{l=0}^p G_{pl} |f^{(l)}(x)|^2 + 2 \sum_{0 \leq k < j \leq p} H_{pkj} \operatorname{Re}(r_{pkj} f^{(k)}(x) \overline{f^{(j)}(x)}) \quad (21)$$

holds for arbitrary $p \in \mathbb{N}_0$, where r_{pkj} , G_{pl} , H_{pkj} are defined in (18) and (19).

Proposition 3. If $h, \omega, \omega_p: \mathbb{R} \rightarrow \mathbb{R}$ are real signals, $v = p-2q$, $0 \leq q \leq \frac{p}{2}$, if

$$\sum_{r=0}^{v-1} (-1)^r \lim_{|x| \rightarrow \infty} \omega_p^{(r)}(x) h^{(v-r-1)}(x) = 0, \quad (22)$$

then

$$\int_{\mathbb{R}} \omega_p(x) h^{(v)}(x) dx = \sum_{r=0}^{v-1} (-1)^r \omega_p^{(r)}(x) h^{(v-r-1)}(x)$$

$$+ (-1)^v \int_{\mathbb{R}} \omega_p^{(v)}(x) h(x) dx \quad (23)$$

and

$$\int_{\mathbb{R}} \omega_p(x) h^{(v)}(x) dx = (-1)^v \int_{\mathbb{R}} \omega_p^{(v)}(x) h(x) dx \quad (24)$$

holds for arbitrary $p, v \in \mathbb{N}_0$.

Proposition 4. If $f, f_{\tilde{\alpha}}: \mathbb{R} \rightarrow \mathbb{C}$ are complex signals, $f_{\tilde{\alpha}}(x) = e^{\tilde{\alpha}x} f(x)$, $\tilde{\alpha} = -i2\pi\xi_m$, $f, f_{\tilde{\alpha}}^{(p)}$, and $(\xi - \xi_m)F_f$ are all in $L^2(\mathbb{R})$. Then,

$$\sigma_{\xi}^{2p} = \frac{1}{(2\pi)^{2p}} \int_{\mathbb{R}} |f_{\tilde{\alpha}}^{(p)}(x)|^2 dx \quad (25)$$

holds for arbitrary $p \in \mathbb{N}_0$.

3. Plancherel–Parseval–Rayleigh identities for LCT

In order to obtain the weighted uncertainty principle for LCT, we investigate the Plancherel–Parseval–Rayleigh identities related to the LCT in this section. First, we obtain the identity of one dimensional signal associated with the LCT as following Theorem 1.

Theorem 1. If $f(x) \in L^2(\mathbb{R})$ is a complex signal, $g(x) = f(x)e^{i\frac{\alpha}{2b}x^2}$, $g_{\alpha}(x) = e^{\alpha x} g(x)$, $g_{\alpha}^{(p)}(x) = \frac{d^p}{dx^p} g_{\alpha}(x)$, $\alpha = -i\frac{\xi_m}{b}$, $f, L_f^A, (\xi - \xi_m)^p L_f^A \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Then,

$$\sigma_{\xi,A}^{2p} = b^{2p} \int_{\mathbb{R}} |g_{\alpha}^{(p)}(x)|^2 dx \quad (26)$$

holds for arbitrary $p \in \mathbb{N}_0$, and $b > 0$.

Proof. From Definition 1, we have

$$\begin{aligned} \sigma_{\xi,A}^{2p} &= \int_{\mathbb{R}} (\xi - \xi_m)^{2p} \left| \sqrt{\frac{1}{i2\pi b}} \int_{\mathbb{R}} f(x) e^{i\frac{\alpha}{2b}x^2 - i\frac{1}{b}\xi x + i\frac{d}{2b}\xi^2} dx \right|^2 d\xi \\ &= \frac{1}{2\pi b} \int_{\mathbb{R}} (\xi - \xi_m)^{2p} \left| \int_{\mathbb{R}} f(x) e^{i\frac{\alpha}{2b}x^2 - i\frac{1}{b}\xi x} dx \right|^2 d\xi \\ &= \frac{1}{2\pi b} \int_{\mathbb{R}} (\xi - \xi_m)^{2p} \left| F_g \left(\frac{\xi}{2\pi b} \right) \right|^2 d\xi, \end{aligned} \quad (27)$$

utilizing the Proposition 4, we can obtain Theorem 1. $\square$

Theorem 1 is a generalized form of property 4, and we will prove the Theorem 3 using the Theorem 1. Next, before we generalize the results of Theorem 1 to the 2D LCT case, the Plancherel–Parseval–Rayleigh identity for 2D FT is derived, we list it as the following Lemma 2 and omit the proof.

Lemma 2. If $f(x, y) \in L^2(\mathbb{R}^2)$ is a complex signal, $f_{\tilde{\alpha}}(x, y) = e^{\tilde{\alpha}x} f(x, y)$, $f_{\tilde{\beta}}(x, y) = e^{\tilde{\beta}y} f(x, y)$, $\tilde{\alpha} = -i2\pi\xi_m$, $\tilde{\beta} = -i2\pi\eta_m$, $f(x, y), F_f(\xi, \eta), (\xi - \xi_m)^p F_f(\xi, \eta), (\eta - \eta_m)^p F_f(\xi, \eta) \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$. Then,

$$\sigma_{\xi,\eta}^{2p} = \frac{1}{(2\pi)^{2p}} \int_{\mathbb{R}^2} (|f_{\tilde{\alpha}}^{(p)}(x, y)|^2 + |f_{\tilde{\beta}}^{(p)}(x, y)|^2) dx dy \quad (28)$$

holds for arbitrary $p \in \mathbb{N}_0$.

Finally, we obtain the Plancherel–Parseval–Rayleigh identity for 2D LCT as following Theorem 2.

Theorem 2. If $f(x, y) \in L^2(\mathbb{R}^2)$ is a complex signal, $g(x, y) = f(x, y)e^{i(\frac{a_1 x^2}{2b_1} + \frac{a_2 y^2}{2b_2})}$, $g_{\tilde{\alpha}}(x, y) = e^{\tilde{\alpha}x} g(x, y)$, $g_{\tilde{\beta}}(x, y) = e^{\tilde{\beta}y} g(x, y)$, $\tilde{\alpha} = -i\frac{\xi_m}{b_1}$, $\tilde{\beta} = -i\frac{\eta_m}{b_2}$, $f(x, y), F_f^{\tilde{A}, \tilde{B}}(\xi, \eta), (\xi - \xi_m)^p F_f^{\tilde{A}, \tilde{B}}(\xi, \eta)$ and $(\eta - \eta_m)^p F_f^{\tilde{A}, \tilde{B}}(\xi, \eta) \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$. Then,

$$\sigma_{\xi,\eta,\tilde{A},\tilde{B}}^{2p} = b_1^{2p} \int_{\mathbb{R}^2} |(g_{\tilde{\alpha}}(x, y))_x^{(p)}|^2 dx dy + b_2^{2p} \int_{\mathbb{R}^2} |(g_{\tilde{\beta}}(x, y))_y^{(p)}|^2 dx dy$$

holds for any fixed $p \in \mathbb{N}_0$.

Proof. See Appendix A. $\square$

4. Heisenberg–Pauli–Weyl uncertainty principles for LCT

4.1. 2pth weighted uncertainty principles for LCT

The 2pth weighted Heisenberg–Pauli–Weyl uncertainty principles for LCT can be derived as following Theorem 3.

Theorem 3. If $f \in L^2(\mathbb{R})$ is a complex signal, $\omega : \mathbb{R} \rightarrow \mathbb{R}$ is a real weight signal. $L_f^A(\xi)$ and $(\xi - \xi_m)^p L_f^A(\xi) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Then the following inequation holds, as

$$\sigma_{\omega, x}^{2p} \sigma_{\xi, A}^{2p} \geq \frac{b^{2p}}{4} (E_{p, f}^A)^2 \quad (30)$$

The equality holds if

$$f(x) = e^{i \frac{\xi_m}{b} x} e^{-i \frac{a}{2b} x^2} H(x) \quad (31)$$

here

$$H(x) = \sum_{j=0}^{p-1} \chi_j \tilde{x}^j \left[1 + \sum_{m=0}^{\infty} (-1)^{m+1} \frac{(k_p^2 \tilde{x}^{2p})^{m+1}}{p_p^{2p+j} p_p^{4p+j} \dots p_p^{2(m+1)p+j}} \right] \quad (32)$$

and

$$E_{p, f}^A = \sum_{i=0}^{\lfloor \frac{p}{2} \rfloor} D_i \tilde{D}_i, \quad (33)$$

where $k_p \in \mathbb{R}_+$, $\chi_j \in \mathbb{C}$ ($j = 0, 1, 2, \dots, p-1$), D_i is the same as (14), $\tilde{D}_i$ is similar to (15) in which $\mu = 2\pi \xi_m$ is replaced with $\mu = \frac{\xi_m}{b}$, as well as

$$p_p^{(2m+1)p+j} = (2(m+1)p+j)(2(m+1)p+j-1) \dots ((2m+1)p+j+1) \quad (34)$$

denotes permutations for $p \in \mathbb{N}_0$, $m \in \mathbb{N}$.

Proof. See Appendix B. $\square$

Based on the relationship between the LCT and the FRFT, the 2pth weighted Heisenberg–Pauli–Weyl uncertainty principle for FRFT can derived as following Corollary 1.

Corollary 1. If $f \in L^2(\mathbb{R})$ is a complex signal, $\omega : \mathbb{R} \rightarrow \mathbb{R}$ is a real weight signal. We denote $g(x) = f(x) e^{i \frac{x^2}{2} \cot \theta}$, $g_{\hat{\alpha}}(x) = e^{\hat{\alpha} x} g(x)$, $\hat{\alpha} = -i \xi_m \csc \theta$. $F_f^\theta(\xi)$, $(\xi - \xi_m)^p F_f^\theta(\xi) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Then,

$$\sigma_{\omega, x}^{2p} \sigma_{\xi, \theta}^{2p} \geq \frac{\sin^2 \theta}{4} (E_{p, f}^\alpha)^2 \quad (35)$$

holds. The equality holds when

$$f(x) = e^{i \xi_m x \csc \theta} e^{-i \frac{1}{2} x^2 \cot \theta} H(x) \quad (36)$$

where $H(x)$ is the same as in (32). And

$$E_{p, f}^\alpha = \sum_{i=0}^{\lfloor \frac{p}{2} \rfloor} D_i \hat{D}_i, \quad (37)$$

holds for $p \in \mathbb{N}_0$. $\hat{D}_i$ is similar to (15) in which $\mu = 2\pi \xi_m$ is replaced with $\mu = \frac{\xi_m \csc \theta}{2\pi}$.

When matricial parameter $a = 0$, $b = \frac{1}{2\pi}$, $c = -2\pi$, $d = 0$, the result in (30) reduces to the well known results of 2pth weighted Heisenberg–Pauli–Weyl uncertainty principles in [49].

(29)

Corollary 2. Under the assumptions of Theorem 3, Corollary 1 and Definition 1, when $p = 1$, we can easily obtain the 2th weighted moment of the Heisenberg–Pauli–Weyl uncertainty principles for the LCT, FRFT and FT as

$$\sigma_{\omega, x}^2 \sigma_{\xi, A}^2 \geq \frac{b^2}{4} E_{1, f}^2, \quad (38)$$

$$\sigma_{\omega, x}^2 \sigma_{\xi, \theta}^2 \geq \frac{\sin^2 \theta}{4} E_{1, f}^2, \quad (39)$$

and

$$\sigma_{\omega, x}^2 \sigma_{\xi}^{2p} \geq \frac{1}{16\pi^2} E_{1, f}^2, \quad (40)$$

respectively. The equalities in (38–40) hold when

$$f(x) = c_0 e^{-c_p (x-x_m)^2} e^{i(\frac{\xi_m}{b} x - \frac{a}{2b} x^2)}, \quad (41)$$

$$f(x) = c_0 e^{-c_p (x-x_m)^2} e^{i(x \xi_m \csc \theta - \frac{1}{2} x^2 \cot \theta)}, \quad (42)$$

and

$$f(x) = c_0 e^{-c_p (x-x_m)^2} e^{i 2\pi x \xi_m}, \quad (43)$$

respectively, $c_0, x_m, \xi_m \in \mathbb{R}$, $c_p \in \mathbb{R}_+$ and $p \in \mathbb{N}_0$. with

$$E_{1, f} = - \int_{\mathbb{R}} \tilde{\omega}'(x) |f(x)|^2 dx. \quad (44)$$

Corollary 3. Under the assumptions of Theorem 3, Corollary 1, 2 and Definition 1, when $\omega(x) = 1$ and $\|f\|_2 = 1$, we can easily obtain the 2th Heisenberg–Pauli–Weyl uncertainty principles for the LCT, FRFT and FT as

$$\sigma_x^2 \sigma_{\xi, A}^2 \geq \frac{b^2}{4}, \quad (45)$$

$$\sigma_x^2 \sigma_{\xi, \theta}^2 \geq \frac{\sin^2 \theta}{4}, \quad (46)$$

and

$$\sigma_x^2 \sigma_{\xi}^2 \geq \frac{1}{16\pi^2}, \quad (47)$$

respectively, with

$$E_{1, f} = - \int_{\mathbb{R}} |f(x)|^2 dx = -1. \quad (48)$$

It should be noted that these results are the well known Heisenberg uncertainty principles for LCT derived in [16,18,21], FRFT in [53] and FT in [1,2,3,4,5], respectively. Table 4 summarizes the weighted Heisenberg–Pauli–Weyl uncertainty principles of the LCT and its special cases. It is shown in Table 4 that the classical uncertainty principles can be looked as the special cases of the results derived in this paper.

4.2. 2pth weighted uncertainty principles for 2D LCT

Similar with the derivation of the weighted uncertainty principles for signal in LCT domain, we will investigate the 2pth weighted Heisenberg–Pauli–Weyl uncertainty principles for 2D signals in the LCT domain in this subsection, and the main result can be represented as the following Theorem 4.

Theorem 4. Assume that $f(x, y) \in L^2(\mathbb{R}^2)$ is a complex signal, $g_{\hat{\alpha}}(x, y)$, $g_{\hat{\beta}}(x, y)$, $F_f^{\hat{A}, \hat{B}}(\xi, \eta)$, $(\xi - \xi_m)^p F_f^{\hat{A}, \hat{B}}(\xi, \eta)$, $(\eta - \eta_m)^p F_f^{\hat{A}, \hat{B}}(\xi, \eta) \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$. Then,

$$\sigma_{\omega, x, y}^{2p} \sigma_{\xi, \eta, \hat{A}, \hat{B}}^{2p} \geq \frac{b_1^{2p}}{4} (E_{1, x}^2 + E_{2, x}^2) + \frac{b_2^{2p}}{4} (E_{1, y}^2 + E_{2, y}^2) \quad (49)$$

Table 4
Weighted Heisenberg–Pauli–Weyl uncertainty principles for LCT and its special cases.

Order and moment	WHPWUP for LCT	WHPWUP for FRFT	WHPWUP for FT
2pth weighted moment	$\sigma_{\omega,x}^2 \sigma_{\xi,A}^2 \geq \frac{b^2}{4} E_{p,f}^2$ (Eq. (30))	$\sigma_{\omega,x}^2 \sigma_{\xi,\theta}^2 \geq \frac{\sin^2 \theta}{4} E_{p,f}^2$ (Eq. (35))	$\sigma_{\omega,x}^2 \sigma_{\xi}^2 \geq \frac{1}{2^{2(p+1)} \pi^{2p}} E_{p,f}^2$ ([49])
2th weighted moment ($p = 1$)	$\sigma_{\omega,x}^2 \sigma_{\xi,A}^2 \geq \frac{b^2}{4} E_{1,f}^2$ (Eq. (38))	$\sigma_{\omega,x}^2 \sigma_{\xi,\theta}^2 \geq \frac{\sin^2 \theta}{4} E_{1,f}^2$ (Eq. (39))	$\sigma_{\omega,x}^2 \sigma_{\xi}^2 \geq \frac{1}{16\pi^2} E_{1,f}^2$ (Eq. (40))
2th moment ($p = 1$, $w = 1$, $\ f\ _2 = 1$)	$\sigma_x^2 \sigma_{\xi,A}^2 \geq \frac{b^2}{4}$ ([16,18,21])	$\sigma_x^2 \sigma_{\xi,\theta}^2 \geq \frac{\sin^2 \theta}{4}$ ([53])	$\sigma_x^2 \sigma_{\xi}^2 \geq \frac{1}{16\pi^2}$ ([1–5])

Note: WHPWUP denotes the Weighted Heisenberg–Pauli–Weyl uncertainty principles.

holds, the equality holds if

$$f(x, y) = ce^{i(\frac{\xi_m}{b_1}x + \frac{\eta_m}{b_2}y)} e^{-i(\frac{a_1 x^2}{2b_1} + \frac{a_2 y^2}{2b_2})} H(x, y) \quad (50)$$

holds, and

$$\begin{aligned} E_{1,x} &= \sum_{i=0}^{\lfloor \frac{p}{2} \rfloor} D_i \bar{D}_{1,x,i}, \quad E_{1,y} = \sum_{i=0}^{\lfloor \frac{p}{2} \rfloor} D_i \bar{D}_{1,y,i}, \\ E_{2,x} &= \sum_{i=0}^{\lfloor \frac{p}{2} \rfloor} D_i \bar{D}_{2,x,i}, \quad E_{2,y} = \sum_{i=0}^{\lfloor \frac{p}{2} \rfloor} D_i \bar{D}_{2,y,i}, \end{aligned} \quad (51)$$

where $|E_{1,x}|$, $|E_{1,y}|$, $|E_{2,x}|$ and $|E_{2,y}| < \infty$ hold for $p \in \mathbb{N}_0$.

Proof. See Appendix C. $\square$

This theorem reflects the weighted Heisenberg–Pauli–Weyl uncertainty principles associated with the 2D LCT, similar with the one dimensional case, we can obtain the following Corollaries easily about the related results for FRFT and FT.

Corollary 4. Under the assumptions of Theorem 4 and Definition 2, we can obtain the 2pth weighted moment of the Heisenberg–Pauli–Weyl uncertainty principles for the 2D FRFT and 2D FT as

$$\sigma_{\omega,x,y}^2 \sigma_{\xi,\eta,\alpha,\beta}^2 \geq \frac{\sin^2 \alpha}{4} (E_{1,x}^2 + E_{2,x}^2) + \frac{\sin^2 \beta}{4} (E_{1,y}^2 + E_{2,y}^2) \quad (52)$$

and

$$\sigma_{\omega,x,y}^2 \sigma_{\xi,\eta}^2 \geq \frac{1}{2^{2(p+1)} \pi^{2p}} (E_{1,x}^2 + E_{2,x}^2) + \frac{1}{2^{2(p+1)} \pi^{2p}} (E_{1,y}^2 + E_{2,y}^2), \quad (53)$$

respectively.

Corollary 5. Under the assumptions of Theorem 4, we can obtain the 2th weighted moment of the Heisenberg–Pauli–Weyl uncertainty principles for 2D LCT, 2D FRFT and 2D FT as following

$$\sigma_{\omega,x,y}^2 \sigma_{\xi,\eta,\tilde{A},\tilde{B}}^2 \geq \frac{b_1^2}{4} (E_{1,x}^2 + E_{2,x}^2) + \frac{b_2^2}{4} (E_{1,y}^2 + E_{2,y}^2), \quad (54)$$

$$\sigma_{\omega,x,y}^2 \sigma_{\xi,\eta,\alpha,\beta}^2 \geq \frac{\sin^2 \alpha}{4} (E_{1,x}^2 + E_{2,x}^2) + \frac{\sin^2 \beta}{4} (E_{1,y}^2 + E_{2,y}^2), \quad (55)$$

and

$$\sigma_{\omega,x,y}^2 \sigma_{\xi,\eta}^2 \geq \frac{1}{16\pi^2} (E_{1,x}^2 + E_{2,x}^2 + E_{1,y}^2 + E_{2,y}^2), \quad (56)$$

respectively. Equalities (54), (55) and (56) hold if

$$f(x, y) = ce^{i(\frac{\xi_m}{b_1}x + \frac{\eta_m}{b_2}y)} e^{-i(\frac{a_1 x^2}{2b_1} + \frac{a_2 y^2}{2b_2})} e^{-c_p[(x-x_m)^2 + (y-y_m)^2]},$$

$$f(x, y) = ce^{i(\xi_m x \csc \theta_1 + \eta_m y \csc \theta_2)} e^{-i \frac{x^2 \cot \theta_1 + y^2 \cot \theta_2}{2}} e^{-c_p[(x-x_m)^2 + (y-y_m)^2]},$$

and

$$f(x, y) = ce^{i(2\pi \xi_m x + 2\pi \eta_m y)} \cdot e^{-c_p[(x-x_m)^2 + (y-y_m)^2]},$$

respectively.

Corollary 6. Under the assumptions of Theorem 4, Corollary 4, 5 and Definition 2, when $p = 1$, $\omega(x, y) = 1$ and $\|f(x, y)\|_2 = 1$, we can

obtain the 2th Heisenberg–Pauli–Weyl uncertainty principles for 2D LCT, 2D FRFT and 2D FT as

$$\sigma_{x,y}^2 \sigma_{\xi,\eta,\tilde{A},\tilde{B}}^2 \geq \frac{b_1^2 + b_2^2}{4}, \quad (57)$$

$$\sigma_{x,y}^2 \sigma_{\xi,\eta,\alpha,\beta}^2 \geq \frac{\sin^2 \alpha + \sin^2 \beta}{4}, \quad (58)$$

and

$$\sigma_{x,y}^2 \sigma_{\xi,\eta}^2 \geq \frac{1}{8\pi^2}, \quad (59)$$

respectively.

Table 5 summarizes the weighted Heisenberg–Pauli–Weyl uncertainty principles of 2D LCT and its special cases, it is easy to see that the classical results can all be regarded as special cases of the results achieved in Theorem 4 of this paper.

5. Simulations and applications

In order to show the flexibility of derived results in this paper as compared with the classical results, we first discuss the bound of the derive results with the latest uncertainty principle derived in [22] by two examples in subsection A; and then applications of the derived results in time-frequency analysis and signal energy concentration are investigated in subsection B and C, respectively.

5.1. Comparison with the latest results

It is shown in Tables 4 and 5 that the results derived in this paper are more general than the existing results. We use the following two examples to discuss the tightness and flexibility of the derived results.

Example 1. It is shown in (38) that the following signal $f(x)$

$$f(x) = \rho(x) e^{i\varphi(x)} = e^{-\frac{\gamma}{2}x^2} e^{-i\frac{\alpha}{2b}x^2}, \quad \gamma > 0,$$

will make the uncertainty principle inequality to equality as following

$$\sigma_{\omega,x}^2 \sigma_{\xi,A}^2 \geq \frac{b^2}{4} E_{1,f}^2, \quad (60)$$

Assume $\xi_m = 0$, $x_m = 0$, and let the weighted signal $\omega(x) = e^{-\gamma x}$. Then, we have

$$\sigma_{\omega,x}^2 = \int_{\mathbb{R}} x^2 e^{-\gamma(x^2+2x)} dx = e^{\gamma} (1 + \frac{1}{2\gamma}) \sqrt{\frac{\pi}{\gamma}},$$

$$\sigma_{\xi,A}^2 = b^2 \int_{\mathbb{R}} |(e^{i\frac{\alpha}{2b}x^2} f(x))'|^2 d\xi = \frac{1}{2} b^2 \gamma \sqrt{\frac{\pi}{\gamma}},$$

and

$$\frac{b^2}{4} (E_{1,f})^2 = \frac{b^2}{4} \left(\int_{\mathbb{R}} (1 - \gamma x) e^{-\gamma(x^2+x)} dx \right)^2 = \frac{b^2}{4} e^{\frac{\gamma}{2}} (1 + \frac{\gamma}{2})^2 \frac{\pi}{\gamma}.$$

Hence, we can obtain

$$\sigma_{\omega,x}^2 \sigma_{\xi,A}^2 = \frac{b^2}{4} \frac{\pi}{\gamma} e^{\gamma} (1 + 2\gamma),$$

Table 5

Weighted Heisenberg–Pauli–Weyl uncertainty principles for the 2D LCT and its special cases.

Order and moment	WHPWUP for 2D LCT	WHPWUP for 2D FRFT	WHPWUP for 2D FT
2pth weighted moment	$\sigma_{\omega,x,y}^{2p} \sigma_{\xi,\eta,\tilde{A},\tilde{B}}^{2p} \geq \frac{b_1^{2p}}{4} (E_{1,x}^2 + E_{2,x}^2) + \frac{b_2^{2p}}{4} (E_{1,y}^2 + E_{2,y}^2)$ (Eq. (49))	$\sigma_{\omega,x,y}^{2p} \sigma_{\xi,\eta,\theta_1,\theta_2}^{2p} \geq \frac{\sin^{2p} \theta_1}{4} (E_{1,x}^2 + E_{2,x}^2) + \frac{\sin^{2p} \theta_2}{4} (E_{1,y}^2 + E_{2,y}^2)$ (Eq. (52))	$\sigma_{\omega,x,y}^{2p} \sigma_{\xi,\eta}^{2p} \geq \frac{1}{2^{2(p+1)} \pi^{2p}} (E_{1,x}^2 + E_{2,x}^2) + \frac{1}{2^{2(p+1)} \pi^{2p}} (E_{1,y}^2 + E_{2,y}^2)$ (Eq. (53))
2th weighted moment ($p = 1$)	$\sigma_{\omega,x,y}^2 \sigma_{\xi,\eta,\tilde{A},\tilde{B}}^2 \geq \frac{b_1^2}{4} (E_{1,x}^2 + E_{2,x}^2) + \frac{b_2^2}{4} (E_{1,y}^2 + E_{2,y}^2)$ (Eq. (54))	$\sigma_{\omega,x,y}^2 \sigma_{\xi,\eta,\theta_1,\theta_2}^2 \geq \frac{\sin^2 \theta_1}{4} (E_{1,x}^2 + E_{2,x}^2) + \frac{\sin^2 \theta_2}{4} (E_{1,y}^2 + E_{2,y}^2)$ (Eq. (55))	$\sigma_{\omega,x,y}^2 \sigma_{\xi,\eta}^2 \geq \frac{1}{16\pi^2} (E_{1,x}^2 + E_{2,x}^2) + \frac{1}{16\pi^2} (E_{1,y}^2 + E_{2,y}^2)$ (Eq. (56))
2th moment ($p = 1$ $w = 1, \ f\ _2 = 1$)	$\sigma_{x,y}^2 \sigma_{\xi,\eta,\tilde{A},\tilde{B}}^2 \geq \frac{b_1^2 + b_2^2}{4}$ (Eq. (57))	$\sigma_{x,y}^2 \sigma_{\xi,\eta,\theta_1,\theta_2}^2 \geq \frac{\sin^2 \theta_1 + \sin^2 \theta_2}{4}$ (Eq. (58))	$\sigma_{x,y}^2 \sigma_{\xi,\eta}^2 \geq \frac{1}{8\pi^2}$ (Eq. (59))

Note: WHPWUP denotes the Weighted Heisenberg–Pauli–Weyl uncertainty principles.

and

$$\frac{b^2}{4} \frac{\pi}{\gamma} e^\gamma (1 + 2\gamma) \geq \frac{b^2}{4} e^{\frac{\gamma}{2}} (1 + \frac{\gamma}{2})^2 \frac{\pi}{\gamma} > \frac{b^2}{4},$$

where $b^2 (\frac{1}{4} + \text{COV}_f^2 - \text{Cov}_f^2) + (a\sigma_x^2 + b\text{Cov}_f)^2 = \frac{b^2}{4}$, under the condition that $\gamma > 0$, we conclude that

$$\frac{b^2}{4} (E_{1,f}^2)^2 > b^2 (\frac{1}{4} + \text{COV}_f^2 - \text{Cov}_f^2) + (a\sigma_x^2 + b\text{Cov}_f)^2. \quad (61)$$

where

$$\text{Cov}_f = \int_{\mathbb{R}} x \varphi'(x) |f(x)|^2 dx - \langle x \rangle_f \langle \xi \rangle_f$$

$$\text{COV}_f = \int_{\mathbb{R}} |(x - \langle x \rangle_f)(\varphi'(x) - \langle \xi \rangle_f)| \rho(x)^2 dx$$

and $\langle x \rangle_f = \int_{\mathbb{R}} x |f(x)|^2 dx$, $\langle \xi \rangle_f = \int_{\mathbb{R}} \xi |F_f(\xi)|^2 d\xi$. The right side of (61) is the uncertainty principle derived in [22] (Corollary 3.4), it shows that the lower bounds of the result presented in (38) are strictly stronger than that of [22].

Example 2. We use the same example as in [22] (Example 3.9) to investigate the differences of the results in this paper and in [22]. The signal is rewritten as follows

$$f(x) = \rho(x) e^{i\varphi(x)} = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}} e^{-\frac{\alpha}{2}x^2} e^{i(\frac{1}{2}x^2 - \frac{\alpha}{6}x^4)}, \quad 0 < \alpha \leq 1,$$

where $\rho(x) = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}} e^{-\frac{\alpha}{2}x^2}$, $\varphi(x) = \frac{1}{2}x^2 - \frac{\alpha}{6}x^4$. We also assume $\omega(x) = e^{-\alpha x}$, $\xi_m = 0$, $x_m = 0$. Hence, from (26), we have

$$\sigma_{\omega,x}^2 = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{2}} \int_{\mathbb{R}} x^2 e^{-\alpha(x^2+2x)} dx = e^\alpha \left(1 + \frac{1}{2\alpha}\right),$$

and

$$\sigma_{\xi,A}^2 = b^2 \int_{\mathbb{R}} |g'(x)|^2 dx = \frac{b^2}{2\alpha} \left(\alpha^2 + \frac{a^2}{b^2} + \frac{2}{3}\right),$$

Hence, one finds that

$$\sigma_{\omega,x}^2 \sigma_{\xi,A}^2 = \frac{b^2 e^\alpha (1 + 2\alpha)}{4\alpha^2} \left(\alpha^2 + \frac{a^2}{b^2} + \frac{2}{3}\right),$$

Moreover, from (44), we can obtain

$$\frac{b^2}{4} E_{1,f}^2 = \frac{b^2}{4} \left(\left(\frac{\alpha}{\pi}\right)^{\frac{1}{2}} \int_{\mathbb{R}} (1 - \alpha x) e^{-\alpha(x^2+x)} dx\right)^2 = \frac{b^2}{4} e^{\frac{\alpha}{2}} \left(1 + \frac{\alpha}{2}\right)^2,$$

On the other hand, based on the results proposed in [22], we have

$$b^2 \left(\frac{1}{4} + \text{COV}_f^2 - \text{Cov}_f^2\right) + (a\sigma_x^2 + b\text{Cov}_f)^2 = \frac{b^2}{4} + \frac{6b^2}{\pi e^3 \alpha^2} + \frac{a^2}{4\alpha^2},$$

It can be computed that

$$\frac{\sigma_{\omega,x}^2 \sigma_{\xi,A}^2}{\frac{b^2}{4} E_{1,f}^2} = \frac{e^{\frac{\alpha}{2}} (1 + 2\alpha) \left(\alpha^2 + \frac{a^2}{b^2} + \frac{2}{3}\right)}{\alpha^2 \left(1 + \frac{\alpha}{2}\right)^2} > 1,$$

and

$$\begin{aligned} \frac{b^2}{4} E_{1,f}^2 - \left\{ b^2 \left(\frac{1}{4} + \text{COV}_f^2 - \text{Cov}_f^2\right) + (a\sigma_x^2 + b\text{Cov}_f)^2 \right\} \\ = \frac{b^2}{4} \left(e^{\frac{\alpha}{2}} \left(1 + \frac{\alpha}{2}\right)^2 - \left(\frac{24}{\pi e^3} + \frac{a^2}{b^2} \right) \frac{1}{\alpha^2} - 1 \right) > 0. \end{aligned}$$

For any arbitrary but fixed $a, b \in \mathbb{R}$, and $0 < \alpha \leq 1$. Hence, we can conclude that

$$\sigma_{\omega,x}^2 \sigma_{\xi,A}^2 > \frac{b^2}{4} E_{1,f}^2 > b^2 \left(\frac{1}{4} + \text{COV}_f^2 - \text{Cov}_f^2\right) + (a\sigma_x^2 + b\text{Cov}_f)^2.$$

Examples 1 and 2 show that the lower bound of our new weighted moment Heisenberg–Pauli–Weyl uncertainty principle is tighter than that of the latest one in [22] as a result of controlling the weighted coefficient of the weighted signal $\omega(x)$. In fact, if we let $\alpha = 1$, $a = 1$, $b = -1$, $c = 0$, $d = 1$ in the above equations, we obtain

$$\sigma_{\omega,x}^2 \sigma_{\xi,A}^2 = 5.43656,$$

$$\frac{b^2}{4} E_{1,f}^2 = 0.92740540,$$

$$b^2 \left(\frac{1}{4} + \text{COV}_f^2 - \text{Cov}_f^2\right) + (a\sigma_x^2 + b\text{Cov}_f)^2 = 0.59508629.$$

This numerical result also verify the correctness of the derived results of this paper.

5.2. Time-frequency bandwidth analysis

In time-frequency analysis, the time duration and spectral bandwidth correspond to the time and frequency resolution, respectively [54]. In this case, the time and frequency resolutions cannot be increased at the same time due to the uncertainty principle. However, we can see from the derived results in (30) that the product of the time-frequency bandwidth related to the LCT parameter b and the weight signal $w(x)$. This fact provides us a way of adjusting the signal bandwidth by choosing the suitable weight signal $w(x)$.

In order to show how to adjust the product of time duration and spectral bandwidth base on the derived weighted Heisenberg–Pauli–Weyl uncertainty principle, we choose the result in (38) as an example, it is shown in (38) that the following weighted uncertainty principle holds.

$$\sigma_{\omega,x}^2 \sigma_{\xi,A}^2 \geq \frac{b^2}{4} E_{1,f}^2,$$

with

$$E_{1,f} = - \int_{\mathbb{R}} \tilde{\omega}'(x) |f(x)|^2 dx,$$

The above equation can be rearranged as

$$\sigma_{\xi,A}^2 \geq \frac{\frac{b^2}{4} E_{1,f}^2}{\sigma_{\omega,x}^2}, \quad (62)$$

From right side of (62), we know that frequency bandwidth of a signal in LCT domain is related to parameters b and weight signal $\omega(x)$. This indicates that the spread of $\sigma_{\xi,A}^2$ can be adjusted by choosing a suitable weight signal $\omega(x)$.

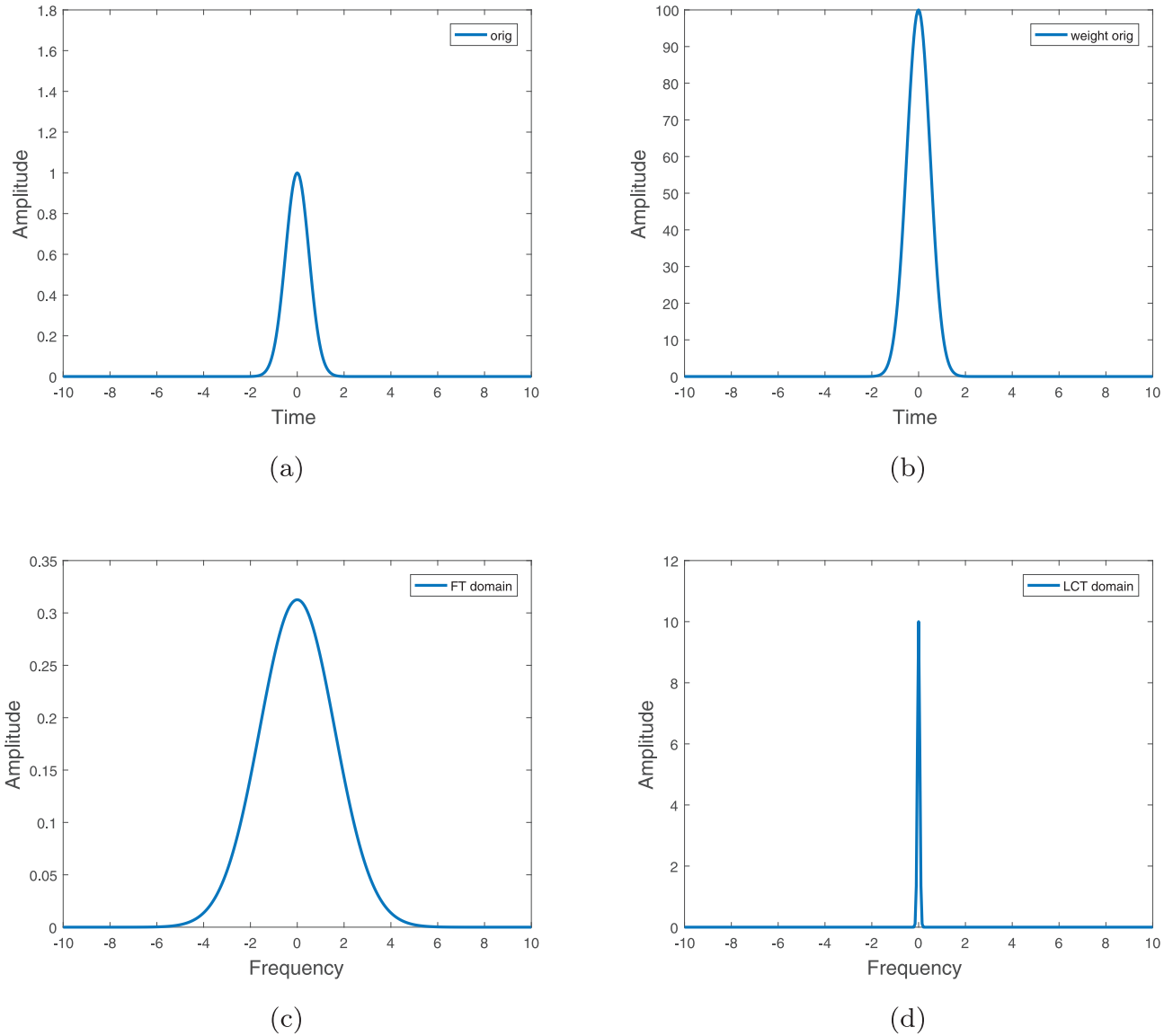


Fig. 1. A signal that can achieve the lower bound with the parameters in (72). (a) energy density of $f(x)$; (b) energy density of weighted $f(x)$; (c) energy density of $F_f(\xi)$; (d) energy density of $L_f^1(\xi)$.

For example, if we choose a signal $f(x)$ as following

$$f(x) = e^{-\frac{x^2}{4} + i\xi x}, \quad (63)$$

with $x_m = 0$, $\omega(x) = e^{-\alpha x^2}$, $\alpha > 0$, we find that

$$\begin{aligned} E_{1,f} &= - \int_{\mathbb{R}} \tilde{\omega}'(x) |f(x)|^2 dx \\ &= \int_{\mathbb{R}} (1 - 2\alpha x^2) e^{-(\alpha + \frac{1}{2})x^2} dx = -\frac{\sqrt{\pi}}{2(\alpha + \frac{1}{2})^{\frac{3}{2}}} \end{aligned} \quad (64)$$

$$\sigma_{\omega,x}^2 = \int_{\mathbb{R}} x^2 e^{-(2\alpha + \frac{1}{2})x^2} dx = \frac{\sqrt{2\pi}}{(1 + 4\alpha)^{\frac{3}{2}}} \quad (65)$$

and

$$\sigma_x^2 = \int_{\mathbb{R}} x^2 e^{-\frac{1}{2}x^2} dx = \sqrt{2\pi} \quad (66)$$

From (62), (64) and (65), we have the lower bound of spectral bandwidth of weighted Heisenberg–Pauli–Weyl uncertainty principle

of (38) as follows

$$\frac{\frac{b^2}{4} E_{1,f}^2}{\sigma_{\omega,x}^2} = \frac{b^2}{4\sqrt{2\pi}} \frac{\pi(1 + 4\alpha)^{\frac{3}{2}}}{4(\alpha + \frac{1}{2})^3} \quad (67)$$

From (45) and (66), we obtain the lower bound of spectral bandwidth of uncertainty principle of (45) as follows

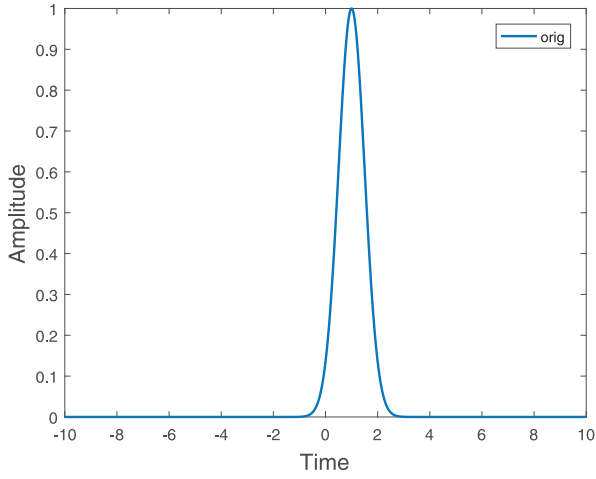
$$\frac{\frac{b^2}{4}}{\sigma_x^2} = \frac{b^2}{4\sqrt{2\pi}} \quad (68)$$

We can find that there are two parameters in (67) rather than one parameter in (68), which indicate that spread of (67) is more flexible than that of (68). For example, if we fix the parameter b , and let α of $w(x)$ meet the condition of $0 < \alpha < 2.6333$, then, we can obtain that

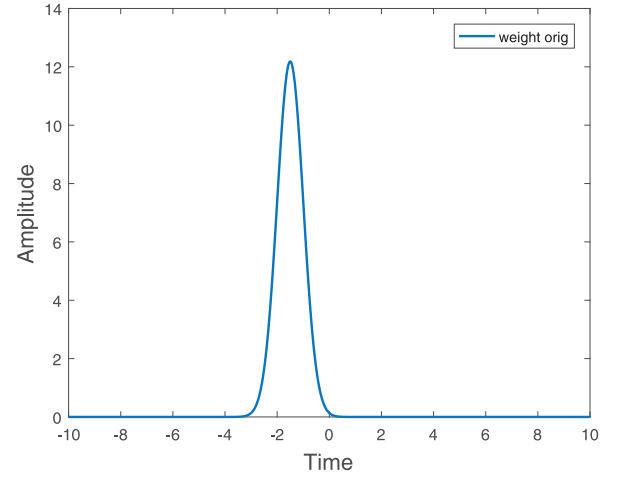
$$\sigma_{\omega,x}^2 < \sigma_x^2 \quad (69)$$

and

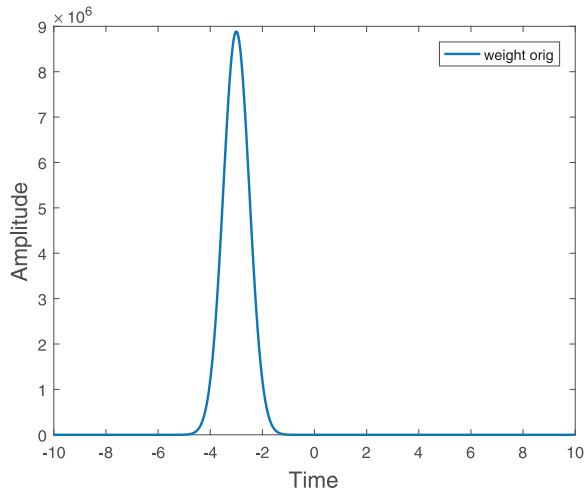
$$\frac{\pi(1 + 4\alpha)^{\frac{3}{2}}}{4(\alpha + \frac{1}{2})^3} > 1 \quad (70)$$



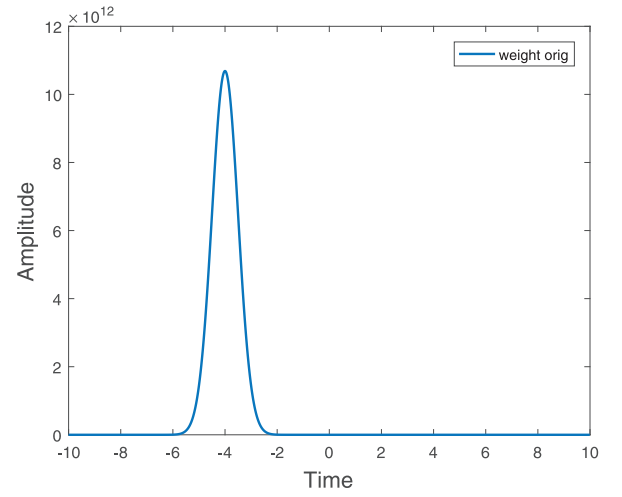
(a)



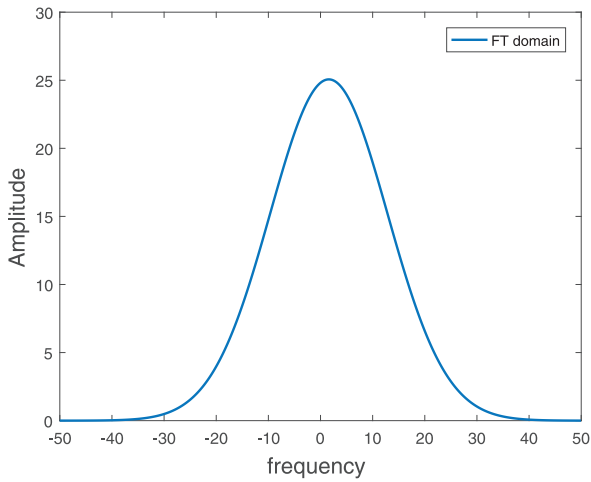
(b)



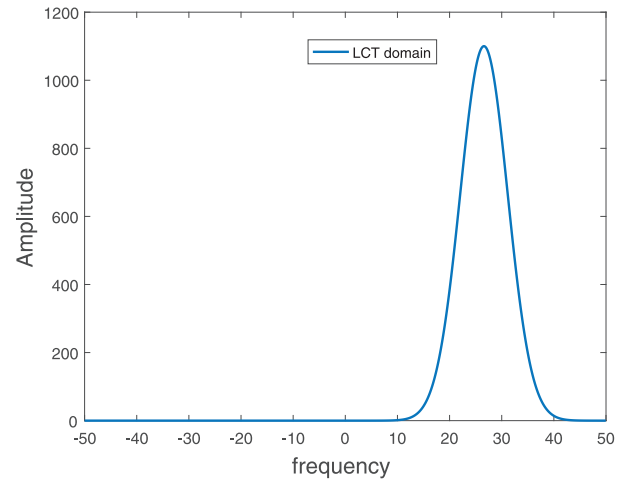
(c)



(d)



(e)



(f)

Fig. 2. A signal that can achieve the lower bound with the parameters in (74). (a) energy density of $g(x)$; (b) energy density of weighted $g(x)$ with $r_3 = 5$; (c) energy density of weighted $g(x)$ with $r_3 = 8$; (d) energy density of weighted $g(x)$ with $r_3 = 10$; (e) energy density of $F_g(\xi)$; (f) energy density of $L_g^A(\xi)$.

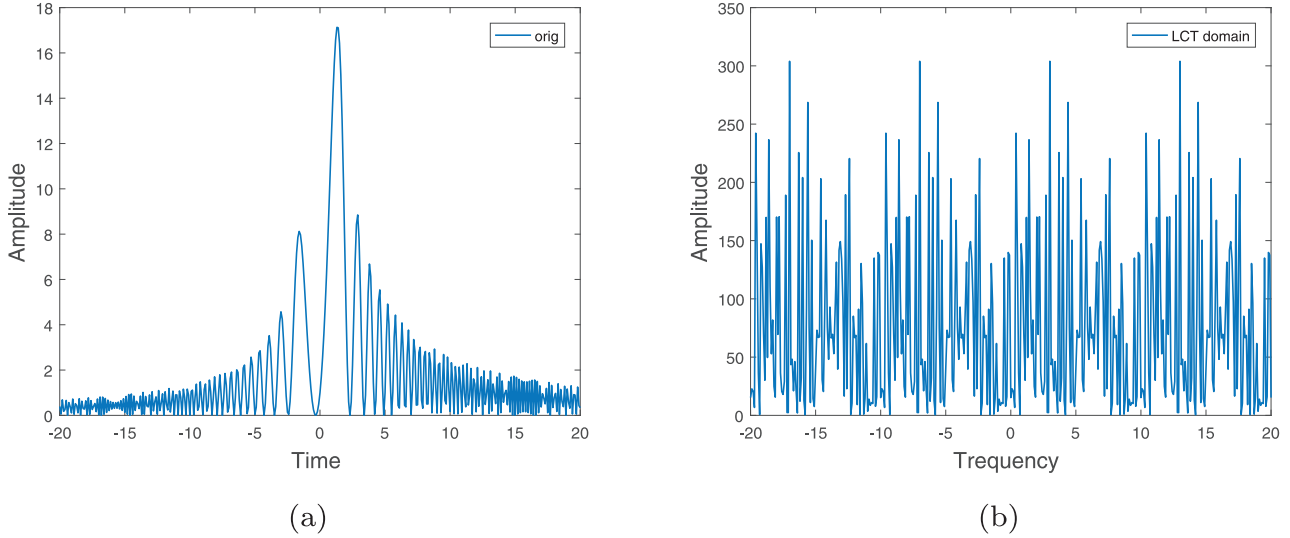


Fig. 3. A signal that can achieve the lower bound with the parameters in (75). (a) energy density of $f(x)$; (b) energy density of $L_f^A(\xi)$.

Multiplying by $\frac{b^2}{4\sqrt{2\pi}}$ on both sides of the above inequality, we have

$$\frac{\frac{b^2}{4} E_{1,f}^2}{\sigma_{\omega,x}^2} > \frac{b^2}{\sigma_x^2} \quad (71)$$

From (69) and (71), we can find that time width is reduced while spectrum bandwidth is improved.

It is shown from these discussions that the lower bound of weighted Heisenberg–Pauli–Weyl uncertainty principle in (38) can be adjusted by choosing the suitable weight signal $w(x)$ (the weight signal $w(x)$ can be arbitrary as long as the condition (11) and (12) are satisfied). From this point of view, the results derived in this paper are more flexible and applicable than the existing ones.

5.3. Simulations

In the field of signal parameter detection and estimation situations, we often need to find a suitable transformation, so that the energy of signal is concentrated in this transform domain. It is concluded from the derived results of this paper that the product of the time duration and the spectral bandwidth is controlled by the parameter b of the LCT and the weight signal $w(x)$. In this subsection, we do some simulations of the signals that attain the proposed uncertainty bounds.

It is shown in (31) of Theorem 3 that the following signal

$$f(x) = e^{-i10x^2} e^{-x^2}$$

that can achieve the lower bound of $\sigma_{\omega,x}^2 \sigma_{\xi,A}^2$, with $\omega(x) = e^{-x}$.

In Fig. 1, we plot the original signal, the weighted signal, the Fourier domain signal and the LCT domain signal in Fig. 1 (a)–(d), respectively. The parameters of LCT are choosing as following

$$a = 1, b = 0.05, c = -2.8, d = 0.86, x_m = 0, \xi_m = 0, \gamma_1 = 1, \gamma_2 = 1 \quad (72)$$

We can conclude from Fig. 1 that the signal's energy is more concentration than the traditional Fourier domain, and it is easy to calculate that

$$\sigma_{\omega,x}^2 \sigma_{\xi,A}^2 = 0.09818949725$$

and

$$\frac{b^2}{4} E_{1,f}^2 = 0.09818213375$$

Which shows that the product of $\sigma_{\omega,x}^2$ and $\sigma_{\xi,A}^2$ is equal to $\frac{b^2}{4} E_{1,f}^2$. This fact fits the derived results in (35) of Theorem 3.

We also show another signal

$$g(x) = r_1 e^{i\left(\frac{\xi_m}{b}(x-x_m) - \frac{a}{2b}(x-x_m)^2\right)} e^{-r_2(x-x_m)^2} \quad (73)$$

that can minimize $\sigma_{\omega,x}^2 \sigma_{\xi,A}^2$, with $\omega(x) = e^{-r_3 x}$. Choose

$$a = 31.85, b = 1, c = 14.92, d = 0.5, x_m = 1, \xi_m = 10, r_1 = 1, r_2 = 1 \quad (74)$$

In Fig. 2, we plot the energy density of original signal in Fig. 2 (a), the energy density of weighted signal with weight $\omega(x) = e^{-5x}$, $\omega(x) = e^{-8x}$ and $\omega(x) = e^{-10x}$ in Fig. 2 (b)–(d), respectively. Energy density of $F_g(\xi)$ and $L_g^A(\xi)$ in Fig. 2 (e) and (f), respectively.

From Fig. 2, we can conclude, although the centers of $g(x)$ and $L_g^A(\xi)$ are no longer at (0,0), the product of $\sigma_{\omega,x}^2 \sigma_{\xi,A}^2$ is still equal to $\frac{b^2}{4} E_{1,f}^2$. We also conclude that the faster the convergence of weight, the more energy concentration of corresponding signal.

We also obtain a signal which can achieve the 4th order weighted Heisenberg–Pauli–Weyl uncertainty principle's lower bound of $\sigma_{\omega,x}^4 \sigma_{\xi,A}^4$ for the LCT, as follows

$$f(x) = e^{i\frac{\xi_m}{b}x} e^{-i\frac{a}{2b}x^2} (C_1 f_1(x) + C_2 f_2(x))$$

where

$$f_1(x) = 1 + \sum_{l=0}^{\infty} (-1)^{m+1} \frac{(k^2 \bar{x}^4)^{l+1}}{3 \cdot 4 \cdots (4l+3) \cdot (4l+4)}$$

$$f_2(x) = \bar{x} \left[1 + \sum_{l=0}^{\infty} (-1)^{m+1} \frac{(k^2 \bar{x}^4)^{l+1}}{4 \cdot 5 \cdots (4l+4) \cdot (4l+5)} \right]$$

and $k \in \mathbb{R} \setminus \{0\}$, $C_1, C_2 \in \mathbb{C}$. With parameters

$$a = 1, b = 0.1, c = -1.4, d = 0.86, k = 1, x_m = 0, \xi_m = 0, \\ C_1 = 1, C_2 = 3, \quad (75)$$

Fig. 3 plots the original signal $f(x)$ and its LCT $L_f^A(\xi)$ in Fig. (a) and (b), respectively. It is shown that if the signal's energy is gathered in the time domain, the energy in the LCT domain is divergent.

6. Conclusions

The Plancherel–Parseval–Rayleigh identities for the LCT is derived, and based on these Plancherel–Parseval–Rayleigh identities, two kinds of $2p$ th weighted Heisenberg–Pauli–Weyl uncertainty principles in LCT domain is obtained in detail. Then, the more general low bounds of weighted version Heisenberg–Pauli–Weyl uncertainty principles for the LCT are achieved. The functions that can achieve the lower bound of the uncertainty principle are also obtained. The classical Heisenberg uncertainty principles in FT domain are shown to be special cases of our achieved results. Examples and simulations are provided to show that our weighted Heisenberg–Pauli–Weyl uncertainty principles are sharper than those in the existing literature. Applications of the derived results in time frequency resolution analysis and signal energy concentrations are also discussed.

Conflict of interest

The authors declared that they have no conflicts of interest to this work.

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Appendix A. Proof of the Theorem 2

Without loss of generality, assume that $b_1 > 0$, $b_2 > 0$. From Lemma 1, we have

$$\sigma_{\xi, \eta, \tilde{A}, \tilde{B}}^{2p} = \int_{\mathbb{R}^2} ((\xi - \xi_m)^{2p} + (\eta - \eta_m)^{2p}) \times \left| \int_{\mathbb{R}^2} f(x, y) K_{\tilde{A}}(\xi, x) K_{\tilde{B}}(\eta, y) dx dy \right|^2 d\xi d\eta. \quad (\text{A.1})$$

Setting $g(x, y) = f(x, y)e^{i(\frac{a_1 x^2}{2b_1} + \frac{a_2 y^2}{2b_2})}$ in (A.1), we find that

$$\sigma_{\xi, \eta, \tilde{A}, \tilde{B}}^{2p} = |C_{\tilde{A}} C_{\tilde{B}}|^2 \int_{\mathbb{R}^2} ((\xi - \xi_m)^{2p} + (\eta - \eta_m)^{2p}) \left| \hat{g}\left(\frac{\xi}{2\pi b_1}, \frac{\eta}{2\pi b_2}\right) \right|^2 d\xi d\eta. \quad (\text{A.2})$$

Let $\gamma_m = \frac{\xi_m}{2\pi b_1}$ and $\delta_m = \frac{\eta_m}{2\pi b_2}$. Based on (A.1), (A.2) and Lemma 2, we can obtain Theorem 2.

Appendix B. Proof of the Theorem 3

Based on Theorem 1, we have

$$\sigma_{\omega, x}^{2p} \sigma_{\xi, A}^{2p} = b^{2p} \left(\int_{\mathbb{R}} \omega_p^2(x) |g_\alpha(x)|^2 dx \right) \times \left(\int_{\mathbb{R}} |g_\alpha^{(p)}(x)|^2 dx \right), \quad (\text{B.1})$$

from (B.1) and the Cauchy–Schwarz inequality, we find that

$$\sigma_{\omega, x}^{2p} \sigma_{\xi, A}^{2p} \geq b^{2p} \left(\int_{\mathbb{R}} |\omega_p(x) g_\alpha(x) g_\alpha^{(p)}(x)| dx \right)^2, \quad (\text{B.2})$$

from (B.2) and the complex inequality, we can obtain

$$\sigma_{\omega, x}^{2p} \sigma_{\xi, A}^{2p} \geq \frac{b^{2p}}{4} \left[\int_{\mathbb{R}} \omega_p(x) \left(g_\alpha(x) \overline{g_\alpha^{(p)}(x)} + \overline{g_\alpha(x)} g_\alpha^{(p)}(x) \right) dx \right]^2, \quad (\text{B.3})$$

from Proposition 1 and (B.3), we find that

$$\sigma_{\omega, x}^{2p} \sigma_{\xi, A}^{2p} \geq \frac{b^{2p}}{4} \left[\int_{\mathbb{R}} \omega_p(x) \left(\sum_{t=0}^{\lfloor \frac{p}{2} \rfloor} D_t \frac{d^{p-2t}}{dx^{p-2t}} |g_\alpha^{(t)}(x)|^2 \right) dx \right]^2, \quad (\text{B.4})$$

From (B.4) and Proposition 2, we have

$$\sigma_{\omega, x}^{2p} \sigma_{\xi, A}^{2p} \geq \frac{b^{2p}}{4} \left[\int_{\mathbb{R}} \omega_p(x) \left(\sum_{t=0}^{\lfloor \frac{p}{2} \rfloor} D_t \frac{d^{p-2t}}{dx^{p-2t}} \left(\sum_{l=0}^t \tilde{G}_{tl} |g^{(l)}(x)|^2 + 2 \sum_{0 \leq k < j \leq t} \tilde{H}_{tkj} \operatorname{Re} \left(r_{tkj} g^{(k)}(x) \overline{g^{(j)}(x)} \right) \right) \right) dx \right]^2, \quad (\text{B.5})$$

From (10)–(13), (B.5) and Proposition 3, we can achieve Theorem 3.

The equality in (30) is upheld by solving the α -differential equation $g_\alpha^{(p)}(x) = -2c_p \tilde{x}^p g_\alpha(x)$ with respect to x , $c_p \in \mathbb{R}_+$. This completes the proof of Theorem 3.

Appendix C. Proof of the Theorem 4

Based on Theorem 2, we can obtain

$$\sigma_{\omega, x, y}^{2p} \sigma_{\xi, \eta, \tilde{A}, \tilde{B}}^{2p} = \int_{\mathbb{R}^2} (\omega_{1,p}^2(x, y) + \omega_{2,p}^2(x, y)) |f(x, y)|^2 dx dy \times \int_{\mathbb{R}^2} (b_1^{2p} |(g_{\tilde{A}}(x, y))_x^{(p)}|^2 + b_2^{2p} |(g_{\tilde{B}}(x, y))_y^{(p)}|^2) dx dy, \quad (\text{C.1})$$

where $g(x, y) = f(x, y)e^{i(\frac{a_1 x^2}{2b_1} + \frac{a_2 y^2}{2b_2})}$, $g_{\tilde{A}}(x, y) = e^{\tilde{\alpha}x} \cdot g(x, y)$, $g_{\tilde{B}}(x, y) = e^{\tilde{\beta}y} g(x, y)$, $\tilde{\alpha} = -i\frac{\xi_m}{b_1}$, $\tilde{\beta} = -i\frac{\eta_m}{b_2}$, $\omega_{1,p}(x, y) = (x - x_m)^p \omega(x, y)$, and $\omega_{2,p}(x, y) = (y - y_m)^p \omega(x, y)$. From (C.1), we find that

$$\sigma_{\omega, x, y}^{2p} \sigma_{\xi, \eta, \tilde{A}, \tilde{B}}^{2p} = b_1^{2p} \left(\int_{\mathbb{R}^2} (\omega_{1,p}(x, y)^2 |g_{\tilde{A}}(x, y)|^2) dx dy \right) \times \left(\int_{\mathbb{R}^2} |(g_{\tilde{A}}(x, y))_x^{(p)}|^2 dx dy \right) + b_1^{2p} \left(\int_{\mathbb{R}^2} (\omega_{2,p}(x, y)^2 |g_{\tilde{A}}(x, y)|^2) dx dy \right) \times \left(\int_{\mathbb{R}^2} |(g_{\tilde{A}}(x, y))_x^{(p)}|^2 dx dy \right) + b_2^{2p} \left(\int_{\mathbb{R}^2} (\omega_{1,p}(x, y)^2 |g_{\tilde{B}}(x, y)|^2) dx dy \right) \times \left(\int_{\mathbb{R}^2} |(g_{\tilde{B}}(x, y))_y^{(p)}|^2 dx dy \right) + b_2^{2p} \left(\int_{\mathbb{R}^2} (\omega_{2,p}(x, y)^2 |g_{\tilde{B}}(x, y)|^2) dx dy \right) \times \left(\int_{\mathbb{R}^2} |(g_{\tilde{B}}(x, y))_y^{(p)}|^2 dx dy \right). \quad (\text{C.2})$$

From (C.2) and the Cauchy–Schwarz inequality, we can obtain

$$\sigma_{\omega, x, y}^{2p} \sigma_{\xi, \eta, \tilde{A}, \tilde{B}}^{2p} \geq b_1^{2p} \left(\int_{\mathbb{R}^2} |\omega_{1,p}(x, y) G_{\tilde{A}}(x, y)| dx dy \right)^2 + b_1^{2p} \left(\int_{\mathbb{R}^2} |\omega_{2,p}(x, y) G_{\tilde{A}}(x, y)| dx dy \right)^2 + b_2^{2p} \left(\int_{\mathbb{R}^2} |\omega_{1,p}(x, y) G_{\tilde{B}}(x, y)| dx dy \right)^2 + b_2^{2p} \left(\int_{\mathbb{R}^2} |\omega_{2,p}(x, y) G_{\tilde{B}}(x, y)| dx dy \right)^2. \quad (\text{C.3})$$

where $G_{\tilde{A}}(x, y) = g_{\tilde{A}}(x, y) (g_{\tilde{A}}(x, y))_x^{(p)}$, $G_{\tilde{B}}(x, y) = g_{\tilde{B}}(x, y) (g_{\tilde{B}}(x, y))_y^{(p)}$. By using (C.3) and the complex inequality, we

obtain

$$\begin{aligned} \hat{M}_p \geq & \frac{b_1^{2p}}{4} \left(\int_{\mathbb{R}^2} \omega_{1,p}(x, y) H_{\tilde{\alpha}}(x, y) dx dy \right)^2 \\ & + \frac{b_1^{2p}}{4} \left(\int_{\mathbb{R}^2} \omega_{2,p}(x, y) H_{\tilde{\alpha}}(x, y) dx dy \right)^2 \\ & + \frac{b_2^{2p}}{4} \left(\int_{\mathbb{R}^2} \omega_{1,p}(x, y) H_{\tilde{\beta}}(x, y) dx dy \right)^2 \\ & + \frac{b_2^{2p}}{4} \left(\int_{\mathbb{R}^2} \omega_{2,p}(x, y) H_{\tilde{\beta}}(x, y) dx dy \right)^2, \end{aligned} \quad (C.4)$$

where $H_{\tilde{\alpha}}(x, y) = \overline{g_{\tilde{\alpha}}(x, y)} (g_{\tilde{\alpha}}(x, y))_x^{(p)} + \overline{g_{\tilde{\alpha}}(x, y)} (g_{\tilde{\alpha}}(x, y))_y^{(p)}$,
 $H_{\tilde{\beta}}(x, y) = \overline{g_{\tilde{\beta}}(x, y)} (g_{\tilde{\beta}}(x, y))_x^{(p)} + \overline{g_{\tilde{\beta}}(x, y)} (g_{\tilde{\beta}}(x, y))_y^{(p)}$. From Proposition 1–3 and the assumptions

$$\begin{aligned} \sum_{r=0}^{p-2q-1} (-1)^r \lim_{|x| \rightarrow \infty} \omega_p(x, y)_x^{(r)} \left(\operatorname{Re}(r_{qkj} f(x, y)_x^{(k)} \overline{(f(x, y))_x^{(j)}}) \right)^{p-2q-1} &= 0, \\ \sum_{r=0}^{p-2q-1} (-1)^r \lim_{|y| \rightarrow \infty} \omega_p(x, y)_y^{(r)} \left(\operatorname{Re}(r_{qkj} f(x, y)_y^{(k)} \overline{(f(x, y))_y^{(j)}}) \right)^{p-2q-1} &= 0, \end{aligned}$$

where $0 \leq k < j \leq q \leq \lfloor \frac{p}{2} \rfloor$, we can obtain the $2p$ th weighted moment Heisenberg–Pauli–Weyl uncertainty principle for the 2D LCT as follows

$$\sigma_{\omega, x, y}^{2p} \sigma_{\xi, \eta, \tilde{A}, \tilde{B}}^{2p} \geq \frac{b_1^{2p}}{4} (E_{1,x}^2 + E_{2,x}^2) + \frac{b_2^{2p}}{4} (E_{1,y}^2 + E_{2,y}^2).$$

where

$$\begin{aligned} \bar{D}_{1,x,s} &= \sum_{l=0}^s \hat{G}_{sl} I_{1,x,sl} + 2 \sum_{0 \leq k < j \leq s} \hat{H}_{skj} I_{1,x,skj}, \\ \bar{D}_{2,x,s} &= \sum_{l=0}^s \hat{G}_{sl} I_{2,x,sl} + 2 \sum_{0 \leq k < j \leq s} \hat{H}_{skj} I_{2,x,skj}, \\ \bar{D}_{1,y,s} &= \sum_{l=0}^s \hat{G}_{sl} I_{1,y,sl} + 2 \sum_{0 \leq k < j \leq s} \hat{H}_{skj} I_{1,y,ikj}, \\ \bar{D}_{2,y,s} &= \sum_{l=0}^s \hat{G}_{sl} I_{2,y,sl} + 2 \sum_{0 \leq k < j \leq s} H_{skj} I_{2,y,skj}. \end{aligned} \quad (C.5)$$

and $E_{1,x}, E_{2,x}, E_{1,y}, E_{2,y}$ are defined in (51). The proof of Theorem 4 is achieved.

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